REPUBLIC OF RWANDA · MINISTRY OF HEALTH · GAKENKE DISTRICT
Ruli District Hospital
Laboratory Department
P.O. Box 1285 Kigali · ruli.hospital@moh.gov.rw · (+250) 0782 000 853 · rulihospital.rw

STANDARD OPERATING PROCEDURE

Venous Blood Sample Collection by Venipuncture

Field	Detail
SOP Number	RliDHL-IH-SOP-001-VERS-001
SOP Title	Venous Blood Sample Collection by Venipuncture
Version	001
Effective Date	14/11/2026
Next Review Date	14/11/2028
Supersedes	— (first issue to the 2026 standard layout)
Document Owner	Kelvin SAMAHANI — Laboratory Manager
Classification	Controlled document — master copy: document control system (RliDHL-ADM-SOP-004)

Approval and Authorisation

Role	Name / Designation	Signature	Date
Prepared by (Author)	**Jean Paul NSENGIYUMVA** **Lab Quality Manager**		
Reviewed by	**Kelvin SAMAHANI** **Laboratory Manager**		
Authorized by	**Dr. Déogratias KANEZA** **Hospital's Director General**		

Review History

Prepared/Reviewed by	Version	Date	Description of Change
J.P. NSENGIYUMVA	001	14/11/2026	First issue to the 2026 standard layout (ISO 15189:2022).

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1. Introduction

Venous blood collection is the most frequently performed pre-examination activity of the laboratory, and pre-examination errors — wrong patient, wrong tube, insufficient volume, haemolysis — are the largest source of unusable specimens. Requests, patient identification, specimen tracking and results are kept in the EMR: no paper request forms accompany specimens inside the hospital. Referred specimens travel with a printed Sample Referral Manifest.

2. Principle of the Test

Blood is drawn from a peripheral vein through a hollow needle into evacuated tubes. The vacuum in each tube draws a fixed volume of blood and mixes it with the additive (anticoagulant, clot activator or preservative) required by the requested examination. Filling the tubes in a fixed order of draw prevents additive carry-over; gentle inversion mixes blood and additive without damaging the cells.

3. Purpose

This SOP standardises the pre-examination steps for the collection of venous blood by peripheral venipuncture so that specimens are collected safely, identified without error, and remain fit for the examinations requested.

4. Scope

This SOP applies to all laboratory personnel — biotechnologists, technicians, interns and students under supervision — who collect venous blood at the main laboratory and at every collection point of Ruli District Hospital. It covers adult and paediatric venipuncture and all tube types listed in 6.3.

It does not cover arterial blood collection, capillary blood sampling, blood culture collection, or blood donation; these are covered by separate SOPs listed in the document control master list.

5. Definitions and Abbreviations

5.1 Definitions

No.	Term	Definition
1	Positive patient identification	Verification of identity with at least two identifiers (full name plus date of birth or patient number) matched to the EMR request before collection.
2	Order of draw	The fixed sequence in which tubes are filled during one venipuncture, preventing additive carry-over between tubes.
3	Haemolysis	Rupture of red blood cells releasing intracellular constituents into serum or plasma; a common cause of specimen rejection.
4	Mislabelled specimen	A specimen whose label does not correctly and uniquely identify the patient; handled as nonconforming work (ISO 15189:2022, 7.5).
5	Referred specimen	A specimen sent to another laboratory for examination; travels with the printed Sample Referral Manifest.

5.2 Abbreviations

No.	Acronym	Full Meaning
1	EHO	Environmental Health Officer
2	EMR	Electronic Medical Record
3	EQA	External Quality Assessment
4	EDTA	Ethylenediaminetetraacetic acid (tube anticoagulant)
5	IPPIS	Integrated Personnel and Payroll Information System
6	LIS	Laboratory Information System
7	NC	Nonconforming (work)
8	NRL	National Reference Laboratory (RBC)
9	PPE	Personal Protective Equipment
10	RBC	Rwanda Biomedical Centre

No.	Acronym	Full Meaning
11	RliDHL	Ruli District Hospital Laboratory
12	SOP	Standard Operating Procedure
13	TAT	Turnaround Time

6. Specimen Requirements

6.1 Request and patient identification

- Accept only requests entered in the EMR by an authorised requester. An oral request is accepted only in an emergency: record who made it and when, and have it entered in the EMR within 24 hours.
- Identify the patient with at least two identifiers — full name plus date of birth or patient number — matched to the EMR request. Scan the wristband where one is worn.
- Each specimen receives its unique laboratory number at reception; the number is scanned onto the tube label and follows the specimen, aliquots and report so that every result traces to one patient.

6.2 Patient preparation and collection

- Confirm the request (examinations, tube types, fasting or timing) before approaching the patient. Explain the procedure, confirm consent, and seat or lie the patient comfortably.
- For fasting specimens, confirm at least 8 hours without food or drink other than water and record the fasting state in the EMR.
- Do not collect from an arm with a running infusion, a fistula, a mastectomy side, or burn/scarred skin unless no alternative exists; if used, note it in the EMR.

6.3 Specimen types, volume, and storage

Tube (cap colour / additive)	Minimum volume	Handling and storage
Blood culture bottles (yellow) — if requested	Per fill mark	Incubate promptly; do not refrigerate
Light blue — sodium citrate 3.2%	Fill to the mark (9:1)	Invert gently 8–10 times; analyse within 24 h at 15–25 °C
Red / gold — plain or clot activator	3 mL	Clot 30 min; centrifuge; serum 8 h at 15–25 °C, 72 h at 2–8 °C
Green — lithium heparin	3 mL	Invert gently; centrifuge promptly; as for serum
Lavender — EDTA	1 mL (2 mL paediatric)	Invert gently 8–10 times; do not shake; examine for clots
Grey — fluoride/oxalate	2 mL	Invert gently; glucose preserved in the unseparated tube (24 h)

6.4 Specimen acceptance and rejection criteria

Criterion	Acceptable	Reject / request recollection
Identification	Label complete; matches the EMR request	Unlabelled, mislabelled, or mismatch with the request
Container	Correct tube type; intact and closed	Wrong tube or additive; leaking or broken tube
Volume	At or above the minimum in 6.3	Below minimum; citrate or EDTA not filled to the mark
Quality	Clear serum or plasma; slight haemolysis noted	Gross haemolysis; clots in an EDTA tube; infusion arm
Timing	Within the limits in 6.3; collection time recorded	Limits exceeded; collection time unknown
Transport	Within 2 hours internally; manifest for referrals	Delayed beyond stability; referral without manifest

Record every rejected specimen in the specimen rejection register and inform the requester at once. If a specimen that fails these criteria is irreplaceable or clinically critical, the Laboratory Manager or designee may authorise testing; the report then states the problem and the caution needed to interpret the result.

7. Responsibilities and Authorisation

Task	Responsible Person(s)
Authorise this SOP and provide the resources needed to apply it	Hospital Director General — Dr. Déogratias KANEZA
Review this SOP at least every two years; ensure users are trained, assessed and authorised; approve recollection exceptions	Laboratory Manager — Kelvin SAMAHANI
Maintain this SOP under document control; monitor indicators, nonconformities and corrective actions; maintain training and authorisation records	Lab Quality Manager — Jean Paul NSENGIYUMVA
Perform venipuncture; identify the patient; label at the bedside; record the collection in the EMR	Authorised Laboratory Biotechnologists
Receive specimens; decide acceptance and rejection (6.4); maintain the specimen rejection register	Laboratory Receptionist / Laboratory Biotechnologist on duty
Manage exposure incidents, sharps and waste; keep PPE and vaccination current	Environmental Health Officer (EHO); all laboratory staff

Authorisation. Only staff trained on this SOP, assessed as competent and authorised by the Laboratory Manager perform venipuncture without supervision. Students and staff in training work under the direct supervision of an authorised person, who checks and countersigns their work. Training, competence assessment and authorisation are recorded in the personnel training file; acknowledgement of this SOP is recorded on the Signature Page (section 21). Competence is reassessed at the interval set in the competence assessment procedure and whenever this SOP changes.

8. Equipment, Reagents, and Consumables

8.1 Equipment

- Vacuum tube holders and safety-engineered needles 21G or 22G; butterfly sets for difficult veins and paediatric patients.
- Disposable single-patient tourniquets.
- Mobile label printer and barcode scanner linked to the EMR.
- Cooled carriers (2–8 °C) and ambient transport boxes where required.

8.2 Reagents, calibrators and controls

Not applicable — this procedure uses no reagents, calibrators or controls. Alcohol swabs and tubes are consumables (8.3).

8.3 Consumables

- Evacuated tubes listed in 6.3, within expiry and stored as the manufacturer directs.
- 70% isopropyl alcohol swabs, dry sterile gauze, adhesive plaster.
- EMR-generated barcode tube labels; biohazard transport bags.
- Puncture-resistant sharps containers; clinical waste bags; 10% sodium hypochlorite; 70% ethanol.

8.4 Calibration and metrological traceability

Not applicable — no measuring equipment is used in this procedure. Centrifuges and analysers used downstream are controlled under the equipment management procedure.

9. Environment and Biosafety Measures

- Treat every specimen as potentially infectious and apply standard precautions.
- Wear a laboratory coat and gloves; change gloves between patients; add eye protection where splashing is possible. Perform hand hygiene before and after every collection.
- Never recap needles. Activate the safety device and discard the assembly in the sharps container at the point of use.
- Hepatitis B vaccination is mandatory for all staff who collect blood.
- After any exposure: wash the area immediately, inform the Laboratory Manager and follow the post-exposure procedure in the Laboratory Safety Manual. Record the event; raise a nonconformity where procedural failure contributed (7.5).

- Clean spills with 10% sodium hypochlorite; decontaminate benches with 70% ethanol. Discard used tubes and materials under the Laboratory Waste Management procedure (waste segregation is supervised by the EHO).
- Keep all patient information confidential (ISO 15189:2022, 4.2).

10. Procedural Steps

10.1 Preparation and patient identification

1. Open the collection worklist on the EMR station or tablet; verify the pending requests.
2. Assemble the tubes in the order of draw (10.2), needle, swabs, gauze, labels and sharps container within reach.
3. Identify the patient with two identifiers matched to the EMR request; scan the wristband where present. Perform hand hygiene and put on gloves.

10.2 Order of draw

When more than one tube is collected during one venipuncture, fill the tubes in the sequence below (consistent with CLSI GP41). Invert each filled tube gently 8 to 10 times; never shake. Fill citrate and EDTA tubes to the mark.

Order	Tube cap colour / additive	Typical examinations
1	Blood culture bottles (yellow)	Blood cultures
2	Light blue — sodium citrate 3.2%	Coagulation (PT, aPTT, INR)
3	Red / gold — plain or clot activator (with gel)	Clinical chemistry, serology, immunology
4	Green — lithium heparin	Plasma chemistry, emergency tests
5	Lavender — EDTA	Haematology (CBC), HbA1c
6	Grey — fluoride / oxalate	Glucose, lactate

10.3 Venipuncture

1. Select the site: prefer the median cubital vein of the forearm. Avoid the areas listed in 6.2.
2. Apply the tourniquet 3 to 4 cm above the site. Do not leave it on for more than 1 minute.
3. Palpate the vein. Clean the site with a 70% isopropyl swab in a widening circle over at least 2 cm and allow to air-dry for at least 30 seconds. Do not touch, blow on or fan the site.
4. Anchor the vein, holding the skin taut. Insert the needle bevel upwards at 15 to 30 degrees in one smooth motion.
5. When blood flows, fill the tubes in the order of draw (10.2). Under-filling or over-filling citrate and EDTA tubes is a cause for rejection.
6. Release the tourniquet before withdrawing the needle. Withdraw, apply dry gauze with firm pressure for 2 to 3 minutes.
7. Activate the needle safety device immediately and discard the assembly in the sharps container at the point of use.
8. If blood does not flow: do not probe more than twice. Release the tourniquet, withdraw, apply pressure, and use a second site or the other arm. Record every difficult draw in the EMR.

10.4 Labelling and identification

- Label every tube at the bedside, in the patient's presence, immediately after filling: scan the wristband, then scan/print the tube label from the EMR. Never pre-label tubes and never label away from the patient.
- Verify each label against the request before leaving the patient. The label carries the patient name, patient number, laboratory number, collection date and time, and the collector's identifier.
- For referred specimens, record the Sample Referral Manifest number on the tube label and in the EMR.
- Corrections are made in the EMR only, traceable to the user account and time; on a printed manifest: single line through, correct entry, initial and date.

10.5 Transport to the laboratory

- Internal transport: no paper accompanies the specimen. Deliver within 2 hours at 15–25 °C, or at 2–8 °C in a cooled carrier where required.

- Do not use the pneumatic tube system for coagulation samples or blood culture bottles unless validated for them.
- Referred specimens: print the Sample Referral Manifest as required by the national referral system; sender and receiving laboratory sign it and a copy is filed with the EMR episode.

10.6 Post-collection

- Record in the EMR: collection time, collector, tube types, fasting state where relevant, and any note (difficult draw, haematoma, infusion arm).
- Dispose of waste and decontaminate the collection area as in section 9.

11. Quality Control and External Quality Assessment

11.1 Quality indicators (internal monitoring)

- Specimen rejection rate: at most 2% per month.
- Haemolysis rate: at most 1% per month.
- Mislabelled specimens: zero — every case is a nonconformity.
- Specimens delivered within 2 hours: at least 95%.
- The Lab Quality Manager extracts these indicators monthly from the EMR/LIS and reviews them with the Laboratory Manager.

11.2 External quality assessment

The laboratory takes part in the pre-analytical or relevant EQA schemes organised or designated by RBC/NRL and acts on unsatisfactory performance.

11.3 Actions when monitoring fails

When an indicator misses its target or an EQA result is unsatisfactory: investigate the cause, define corrective action and evaluate its effectiveness (ISO 15189:2022, 8.7). Where wrong-patient or mislabelled specimens were involved, perform a look-back to the patients affected and inform the Laboratory Manager immediately.

12. Result Interpretation and Validation

Not applicable — this procedure produces no examination result. The decision on specimen fitness is made at reception (6.4). Interpretation and validation of results are covered by the examination SOP concerned.

13. Performance Characteristics and Limitations

- Tourniquet time beyond 1 minute causes haemoconcentration and raises protein and lipid results.
- Squeezing the puncture site causes haemolysis and tissue-fluid contamination.
- Difficult veins and paediatric collection limit volume and quality; record the limitation in the EMR where it may affect a result.
- This procedure controls the pre-examination phase only; method performance is stated in the examination SOP concerned.

14. Reference Intervals, Clinical Decision Limits, and Critical Results

Not applicable — no values are produced by this procedure. Reference intervals, decision limits and critical results are defined in the examination SOPs.

15. Reporting and Release of Results

Not applicable for results — reporting is covered by the examination SOPs. The collection event, specimen status (accepted/rejected) and any recollection request are recorded in the EMR and are traceable to the collector’s user account (ISO 15189:2022, 4.2, 7.4).

16. Specimen Retention and Disposal

- Collected specimens are held and transported under the conditions in 6.3 and 10.5; retention after examination follows the examination SOP concerned.
- Used tubes, swabs and dressings are discarded as infectious waste; sharps go into the puncture-resistant container at the point of use.

17. Risk Management, Nonconforming Work, and Contingency

Risk	Control measure
Wrong-patient identification	Two identifiers at the bedside; wristband scan; label printed and scanned in the patient's presence; never pre-label
Needlestick injury	Safety-engineered devices; no recapping; sharps container at the point of use; hepatitis B vaccination; post-exposure procedure
Haemolysed or clotted specimens	Correct fill volumes; gentle inversion; order of draw; tourniquet under 1 minute; prompt transport
Delayed transport affecting stability	Two-hour delivery rule; scheduled collection runs; cooled carriers
Loss of EMR availability	Downtime procedure below: manual labels double-checked by a second person; back-entry within 24 hours

Nonconforming work (ISO 15189:2022, 7.5): if any step was not followed — wrong tube, missed identification, unlabelled tube — stop, report to the Laboratory Manager, and record the event. The Laboratory Manager decides on the affected specimens, arranges recollection, and determines whether a look-back to released results is needed.

Contingency (ISO 15189:2022, 7.8): when the EMR/LIS is unavailable, use the approved downtime request form and manual tube labels; a second person double-checks every label; keep a written collection log; enter all data into the EMR within 24 hours of restoration. Referral manifests continue to be printed as required by the national referral system.

18. Records

Record	Where kept	Retention
Collection events, labels, collector identity	EMR / LIS (electronic)	As set in the laboratory records policy
Specimen rejection register	EMR / LIS (electronic)	As above
Nonconformity and corrective action records	Quality module (electronic)	As above
Training, competence and authorisation (section 21)	Personnel training file	Duration of employment plus the period required by national rules
Exposure incident reports	Health & Safety file	As above
Sample Referral Manifest (printed)	Referral file; copy with the EMR episode	Per national referral policy

Records are created at the time the activity is performed, are legible, traceable to the person or user account, and are retrievable (ISO 15189:2022, 8.4; RliDHL-ADM-SOP-004).

19. Supporting Documents

- Document and Record Control (RliDHL-ADM-SOP-004).
- Pre-Examination Procedure (RliDHL-ADM-SOP-005).
- Laboratory Safety Manual (post-exposure procedure).
- Laboratory Waste Management procedure.
- Result Reporting and Release procedure.
- Sample Referral procedure and the national Sample Referral Manifest.
- Competence Assessment and Training procedure.

20. References

- ISO 15189:2022 — Medical laboratories: requirements for quality and competence.
- CLSI GP41 — Collection of Diagnostic Venous Blood Specimens.
- CLSI H3 — Procedures for the Handling and Processing of Blood Specimens.
- WHO — Laboratory Quality Management System (LQMS) Handbook.
- Government of Rwanda paperless operations policy (01 August 2026).

21. Signature Page — Training and Competence Acknowledgement (Form F-TRN-001)

I confirm that I have been trained on this SOP, that I have read and understood it, including the safety precautions, and that I agree to perform it as written. I will not perform it without supervision until this record is complete.

S/No	Full Name (print)	IPPIS No.	Designation	Signature	Date
1					
2					
3					
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11					
12					
13					
14					
15					

Trainer / Supervisor: _____ Name (print): _____ Date: _____

Enter the IPPIS number (staff cards are not numbered). If more rows are needed, attach continuation sheet F-TRN-001-2.

22. Amendment Page

Handwritten amendments are authorised by the Laboratory Manager and are entered here word for word before the next revision (ISO 15189:2022, 8.3).

Date	Section	Amendment (word for word)	Reason	Authorised by